


```
In [6]: df.notnull()
```

Out[6]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	True	True	True	True	True	True	True
1	True	True	True	False	True	True	True
2	True	False	True	True	True	False	True
3	True	True	False	True	True	True	True
4	True	True	False	True	True	True	True
5	True	True	True	True	True	True	True
6	True	True	True	False	True	True	True
7	True	True	False	True	True	True	True
8	False	True	True	True	False	True	True
9	True	True	True	True	True	True	True

```
In [7]: df['Maths'] = df['Maths'].fillna(df['Maths'].mean())
```

```
In [8]: df
```

Out[8]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.000000	77.0	86.0	88.0	Male	2018
1	lmn	88.000000	200.0	NaN	76.0	Female	2019
2	stu	81.111111	88.0	170.0	91.0	NaN	2022
3	xyz	73.000000	NaN	87.0	84.0	Female	2018
4	efg	44.000000	NaN	69.0	96.0	Female	2017
5	klm	88.000000	91.0	91.0	84.0	Male	2023
6	rst	72.000000	78.0	NaN	80.0	Female	2016
7	uvw	84.000000	NaN	88.0	78.0	Male	2020
8	NaN	93.000000	84.0	92.0	NaN	Male	2020
9	def	92.000000	93.0	83.0	93.0	Male	2019

```
In [9]: df['Writing'] = df['Writing'].fillna(df['Writing'].median())
```

```
In [10]: df
```

Out[10]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.000000	77.0	86.0	88.0	Male	2018
1	lmn	88.000000	200.0	87.5	76.0	Female	2019
2	stu	81.111111	88.0	170.0	91.0	NaN	2022
3	xyz	73.000000	NaN	87.0	84.0	Female	2018
4	efg	44.000000	NaN	69.0	96.0	Female	2017
5	klm	88.000000	91.0	91.0	84.0	Male	2023
6	rst	72.000000	78.0	87.5	80.0	Female	2016
7	uvw	84.000000	NaN	88.0	78.0	Male	2020
8	NaN	93.000000	84.0	92.0	NaN	Male	2020
9	def	92.000000	93.0	83.0	93.0	Male	2019

In [11]:

```
df['Placement'] = df['Placement'].fillna(df['Placement'].mode())
```

In [12]:

```
df
```

Out[12]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.000000	77.0	86.0	88.0	Male	2018
1	lmn	88.000000	200.0	87.5	76.0	Female	2019
2	stu	81.111111	88.0	170.0	91.0	NaN	2022
3	xyz	73.000000	NaN	87.0	84.0	Female	2018
4	efg	44.000000	NaN	69.0	96.0	Female	2017
5	klm	88.000000	91.0	91.0	84.0	Male	2023
6	rst	72.000000	78.0	87.5	80.0	Female	2016
7	uvw	84.000000	NaN	88.0	78.0	Male	2020
8	NaN	93.000000	84.0	92.0	NaN	Male	2020
9	def	92.000000	93.0	83.0	93.0	Male	2019

In [13]:

```
df['Writing'] = df['Writing'].fillna(df['Writing'].max())
```

In [14]:

```
df
```

Out[14]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.000000	77.0	86.0	88.0	Male	2018
1	lmn	88.000000	200.0	87.5	76.0	Female	2019
2	stu	81.111111	88.0	170.0	91.0	NaN	2022
3	xyz	73.000000	NaN	87.0	84.0	Female	2018
4	efg	44.000000	NaN	69.0	96.0	Female	2017
5	klm	88.000000	91.0	91.0	84.0	Male	2023
6	rst	72.000000	78.0	87.5	80.0	Female	2016
7	uvw	84.000000	NaN	88.0	78.0	Male	2020
8	NaN	93.000000	84.0	92.0	NaN	Male	2020
9	def	92.000000	93.0	83.0	93.0	Male	2019

In [15]:

```
df = pd.DataFrame(dic)
```

In [16]:

```
df
```

Out[16]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.0	77.0	86.0	88.0	Male	2018
1	lmn	88.0	200.0	NaN	76.0	Female	2019
2	stu	NaN	88.0	170.0	91.0	NaN	2022
3	xyz	73.0	NaN	87.0	84.0	Female	2018
4	efg	44.0	NaN	69.0	96.0	Female	2017
5	klm	88.0	91.0	91.0	84.0	Male	2023
6	rst	72.0	78.0	NaN	80.0	Female	2016
7	uvw	84.0	NaN	88.0	78.0	Male	2020
8	NaN	93.0	84.0	92.0	NaN	Male	2020
9	def	92.0	93.0	83.0	93.0	Male	2019

In [17]:

```
df.dropna(how='all')
```

Out[17]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.0	77.0	86.0	88.0	Male	2018
1	lmn	88.0	200.0	NaN	76.0	Female	2019
2	stu	NaN	88.0	170.0	91.0	NaN	2022
3	xyz	73.0	NaN	87.0	84.0	Female	2018
4	efg	44.0	NaN	69.0	96.0	Female	2017
5	klm	88.0	91.0	91.0	84.0	Male	2023
6	rst	72.0	78.0	NaN	80.0	Female	2016
7	uvw	84.0	NaN	88.0	78.0	Male	2020
8	NaN	93.0	84.0	92.0	NaN	Male	2020
9	def	92.0	93.0	83.0	93.0	Male	2019

In [18]: `df.dropna(how='any')`

Out[18]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.0	77.0	86.0	88.0	Male	2018
5	klm	88.0	91.0	91.0	84.0	Male	2023
9	def	92.0	93.0	83.0	93.0	Male	2019

In [19]: `df`

Out[19]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.0	77.0	86.0	88.0	Male	2018
1	lmn	88.0	200.0	NaN	76.0	Female	2019
2	stu	NaN	88.0	170.0	91.0	NaN	2022
3	xyz	73.0	NaN	87.0	84.0	Female	2018
4	efg	44.0	NaN	69.0	96.0	Female	2017
5	klm	88.0	91.0	91.0	84.0	Male	2023
6	rst	72.0	78.0	NaN	80.0	Female	2016
7	uvw	84.0	NaN	88.0	78.0	Male	2020
8	NaN	93.0	84.0	92.0	NaN	Male	2020
9	def	92.0	93.0	83.0	93.0	Male	2019

In [20]: `df.dropna(axis=0, how='any')`

Out[20]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.0	77.0	86.0	88.0	Male	2018
5	klm	88.0	91.0	91.0	84.0	Male	2023
9	def	92.0	93.0	83.0	93.0	Male	2019

In [21]:

```
df.dropna(axis=0, how='all')
```

Out[21]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.0	77.0	86.0	88.0	Male	2018
1	lmn	88.0	200.0	NaN	76.0	Female	2019
2	stu	NaN	88.0	170.0	91.0	NaN	2022
3	xyz	73.0	NaN	87.0	84.0	Female	2018
4	efg	44.0	NaN	69.0	96.0	Female	2017
5	klm	88.0	91.0	91.0	84.0	Male	2023
6	rst	72.0	78.0	NaN	80.0	Female	2016
7	uvw	84.0	NaN	88.0	78.0	Male	2020
8	NaN	93.0	84.0	92.0	NaN	Male	2020
9	def	92.0	93.0	83.0	93.0	Male	2019

In [22]:

```
df
```

Out[22]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.0	77.0	86.0	88.0	Male	2018
1	lmn	88.0	200.0	NaN	76.0	Female	2019
2	stu	NaN	88.0	170.0	91.0	NaN	2022
3	xyz	73.0	NaN	87.0	84.0	Female	2018
4	efg	44.0	NaN	69.0	96.0	Female	2017
5	klm	88.0	91.0	91.0	84.0	Male	2023
6	rst	72.0	78.0	NaN	80.0	Female	2016
7	uvw	84.0	NaN	88.0	78.0	Male	2020
8	NaN	93.0	84.0	92.0	NaN	Male	2020
9	def	92.0	93.0	83.0	93.0	Male	2019

In [23]:

```
df.dropna(axis=1, how='any')
```

Out[23]:

Joining Date	
0	2018
1	2019
2	2022
3	2018
4	2017
5	2023
6	2016
7	2020
8	2020
9	2019

```
In [24]: df.replace(to_replace=np.nan, value=60)
```

Out[24]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date
0	abc	96.0	77.0	86.0	88.0	Male	2018
1	lmn	88.0	200.0	60.0	76.0	Female	2019
2	stu	60.0	88.0	170.0	91.0	60	2022
3	xyz	73.0	60.0	87.0	84.0	Female	2018
4	efg	44.0	60.0	69.0	96.0	Female	2017
5	klm	88.0	91.0	91.0	84.0	Male	2023
6	rst	72.0	78.0	60.0	80.0	Female	2016
7	uvw	84.0	60.0	88.0	78.0	Male	2020
8	60	93.0	84.0	92.0	60.0	Male	2020
9	def	92.0	93.0	83.0	93.0	Male	2019

```
In [25]: dic = {
    "Name":["abc","lmn","stu","xyz","efg","klm","rst","uvw",np.nan,"def"],
    "Maths":[96,88,np.nan,73,44,88,72,84,93,92],
    "Reading":[77,200,88,np.nan,np.nan,91,78,np.nan,84,93],
    "Writing":[86,np.nan,170,87,69,91,np.nan,88,92,83],
    "Placement":[88,76,91,84,96,84,80,78,np.nan,93],
    "Gender":["Male","Female",np.nan,"Female","Female","Male","Female", "Male", "Male", "Male"],
    "Joining Date":[2018,2019,2022,2018,2017,2023,2016,2020,2020,2019],
    "Region":["Pune","Mumbai","Delhi",np.nan,"Surat","Solapur","Bengaluru","Kolkata",np.nan,""]
}
```

```
In [26]: df = pd.DataFrame(dic)
```

```
In [27]: df
```

Out[27]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date	Region
0	abc	96.0	77.0	86.0	88.0	Male	2018	Pune
1	lmn	88.0	200.0	NaN	76.0	Female	2019	Mumbai
2	stu	NaN	88.0	170.0	91.0	NaN	2022	Delhi
3	xyz	73.0	NaN	87.0	84.0	Female	2018	NaN
4	efg	44.0	NaN	69.0	96.0	Female	2017	Surat
5	klm	88.0	91.0	91.0	84.0	Male	2023	Solapur
6	rst	72.0	78.0	NaN	80.0	Female	2016	Bengaluru
7	uvw	84.0	NaN	88.0	78.0	Male	2020	Kolkata
8	NaN	93.0	84.0	92.0	NaN	Male	2020	NaN
9	def	92.0	93.0	83.0	93.0	Male	2019	Pune

In [28]:

```
df['Region'].replace(to_replace=np.nan,value='Goa')
```

Out[28]:

```
0      Pune
1    Mumbai
2     Delhi
3        Goa
4     Surat
5    Solapur
6  Bengaluru
7    Kolkata
8        Goa
9      Pune
Name: Region, dtype: object
```

In [29]:

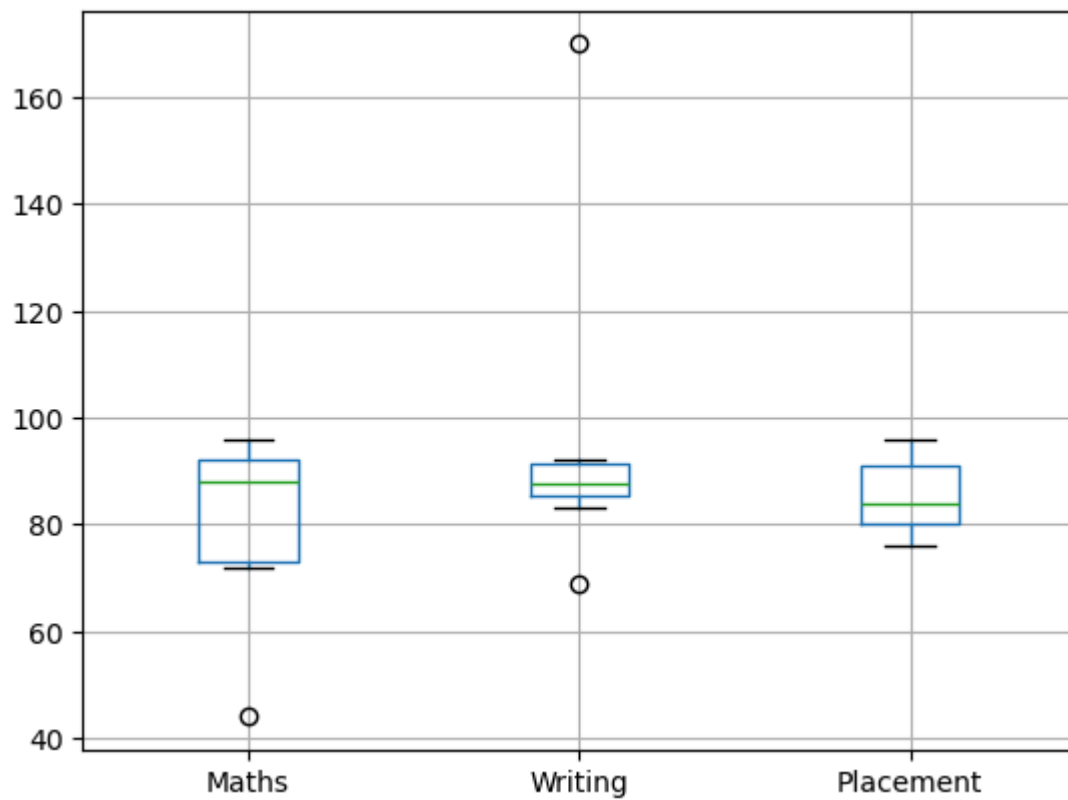
```
col = ['Maths', 'Writing','Placement']
```

In [30]:

```
df.boxplot(col)
```

Out[30]:

```
<Axes: >
```

```
In [31]: df.dropna(inplace=True)
```

```
In [32]: df
```

```
Out[32]:
```

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date	Region
0	abc	96.0	77.0	86.0	88.0	Male	2018	Pune
5	klm	88.0	91.0	91.0	84.0	Male	2023	Solapur
9	def	92.0	93.0	83.0	93.0	Male	2019	Pune

```
In [33]: rscore = df['Writing']
q1 = np.percentile(rscore, 25)
q3 = np.percentile(rscore, 75)
print(q1, q3)
```

```
84.5 88.5
```

```
In [34]: iqr = q3 - q1
print(iqr)
```

```
4.0
```

```
In [35]: lower_bound = q1 - 1.5*iqr
upper_bound = q3 + 1.5*iqr
print(lower_bound, upper_bound)
```

```
78.5 94.5
```

```
In [36]: r_outlier = []
for i in rscore:
    if i < lower_bound or i > upper_bound:
        r_outlier.append(i)
print(r_outlier)
```

```
In [37]: median = np.median(rscore)
median
```

Out[37]: np.float64(86.0)

```
In [38]: df['Writing'] = np.where(df['Writing'] > upper_bound, median, df['Writing'])
df['Writing']
```

Out[38]: 0 86.0
5 91.0
9 83.0
Name: Writing, dtype: float64

```
In [39]: x = df.drop(axis=1, columns=['Region', 'Gender', 'Joining Date'])
x
```

Out[39]:

	Name	Maths	Reading	Writing	Placement
0	abc	96.0	77.0	86.0	88.0
5	klm	88.0	91.0	91.0	84.0
9	def	92.0	93.0	83.0	93.0

```
In [40]: min_max_scaler = preprocessing.MinMaxScaler()
df[['Maths', 'Reading', 'Writing', 'Placement']] = min_max_scaler.fit_transform(df[['Maths',
df
```

Out[40]:

	Name	Maths	Reading	Writing	Placement	Gender	Joining Date	Region
0	abc	1.0	0.000	0.375	0.444444	Male	2018	Pune
5	klm	0.0	0.875	1.000	0.000000	Male	2023	Solapur
9	def	0.5	1.000	0.000	1.000000	Male	2019	Pune